Chapter 6. Explainable Models: Subset Selection, Shrinkage and Dimension Reduction

Lecture 6-3 (best subset, stepwise, C_p/AIC/BIC/CV, ridge, lasso, high-dimensional theory, PCR, PLS)

Contents

1 Three Classes of Methods **2**

2 Subset Selection **2**

2.1 Best subset selection . . . 2

2.2 Stepwise selection . . . 2

2.3 Choosing the optimal model . . . 3

3 Shrinkage Methods **3**

3.1 Ridge regression . . . 3

3.2 Why ridge improves over least squares . . . 3

3.3 The lasso . . . 4

3.4 Constrained formulations and geometry . . . 4

3.5 Dense versus sparse truth . . . 5

3.6 Selecting the tuning parameter . . . 5

4 Theoretical View in High Dimensions **5**

5 Dimension Reduction Methods **6**

5.1 General idea . . . 6

5.2 Principal components regression . . . 6

5.3 Partial least squares . . . 6

5.4 Theoretical comparison . . . 7

6 Selected exercises **7**

1 Three Classes of Methods

- **Subset selection**: least squares on a subset of the p predictors.
- **Shrinkage (regularization)**: shrink all coefficients toward zero; reduces variance, may select variables.
- **Dimension reduction**: least squares on $M < p$ linear projections of the predictors.

2 Subset Selection

Motivation. How to search the 2^p subsets and compare models of different size.

2.1 Best subset selection

Algorithm: best subset selection

1. Let $\mathcal{M}_0$ be the null model (no predictors; predicts the sample mean).
2. For $k = 1, \ldots, p$: fit all $\binom{p}{k}$ models with exactly k predictors and call the best (smallest RSS, or largest R^2) $\mathcal{M}_k$.
3. Select a single best model among $\mathcal{M}_0, \ldots, \mathcal{M}_p$ using a chosen score (CV error, C_p, AIC, BIC, adjusted R^2).

Across sizes RSS always favors the largest model, so step 3 needs a test-error score. 2^p models: infeasible for large p, and the huge search space invites overfitting.

2.2 Stepwise selection

Algorithm: forward stepwise selection

1. Let $\mathcal{M}_0$ be the null model.
2. For $k = 0, \ldots, p-1$: consider the $p-k$ models that add one predictor to $\mathcal{M}_k$; the best of them is $\mathcal{M}_{k+1}$.
3. Select a single best model among $\mathcal{M}_0, \ldots, \mathcal{M}_p$ by CV error, C_p, AIC, BIC or adjusted R^2.

Algorithm: backward elimination

1. Let $\mathcal{M}_p$ be the full model.
2. For $k = p, p-1, \ldots, 1$: consider the k models that drop one predictor from $\mathcal{M}_k$; the best is $\mathcal{M}_{k-1}$.
3. Select a single best model among $\mathcal{M}_0, \ldots, \mathcal{M}_p$.

Both search $1 + \sum_{k=0}^{p-1}(p-k) = 1 + p(p+1)/2$ models (211 versus $2^{20} \approx 10^6$ for $p = 20$), without guarantee: the best two-variable model need not contain the best single predictor. Backward elimination needs $n > p$ ($X^\top X$ invertible); forward selection works for $n < p$ (up to $\mathcal{M}_{n-1}$).

2.3 Choosing the optimal model

Indirect test-error estimates adjust the training error:

$$C_p = \frac{1}{n}(\mathrm{RSS} + 2d\hat{\sigma}^2), \qquad \text{AIC, BIC (Ch. 2)}, \qquad R^2_{\mathrm{adj}},$$

with d the number of parameters and $\hat{\sigma}^2$ estimated from the full model.

Theorem: C_p is unbiased for the in-sample prediction error

For a least-squares fit $\hat{y} = Hy$ with $\mathrm{tr}H = d$, $y = \mu + \varepsilon$, $\mathrm{Var}\varepsilon = \sigma^2 I$, and new responses y^{new} at the same inputs,

$$\mathbb{E}\Big[\frac{1}{n}\|y^{\mathrm{new}} - \hat{y}\|^2\Big] = \mathbb{E}\Big[\frac{1}{n}\|y - \hat{y}\|^2\Big] + \frac{2d\sigma^2}{n}.$$

Proof [Supplement]. Expanding both sides, the difference is $\frac{2}{n}\sum_i \mathrm{Cov}(y_i, \hat{y}_i) = \frac{2}{n}\mathrm{trCov}(y, Hy) = \frac{2}{n}\sigma^2 \mathrm{tr}H = \frac{2d\sigma^2}{n}$. ■

Directly, validation or CV error needs fewer assumptions and no $\hat{\sigma}^2$. Credit data: $\sqrt{\mathrm{BIC}}$, validation and CV error flatten after four to six predictors. **One-standard-error rule**: the smallest model within one SE of the minimum.

3 Shrinkage Methods

Fit all p predictors with coefficients shrunk toward zero, reducing their variance.

3.1 Ridge regression

$$\hat{\beta}^R = \arg\min_{\beta} \Big\{ \sum_{i=1}^{n} \Big(y_i - \beta_0 - \sum_{j=1}^{p} \beta_j x_{ij}\Big)^2 + \lambda \sum_{j=1}^{p} \beta_j^2 \Big\} = \arg\min_{\beta} \{\mathrm{RSS} + \lambda \|\beta\|_2^2\},$$

$\lambda \geq 0$ chosen by CV, intercept not penalized; $\lambda = 0$ is least squares, $\lambda \to \infty$ gives $\hat{\beta}^R \to 0$. For centered data $\hat{\beta}^R = (X^\top X + \lambda I)^{-1} X^\top Y$.

Scaling. Least squares is scale equivariant ($X_j \to cX_j$ gives $\hat{\beta}_j \to \hat{\beta}_j / c$); ridge is not. Standardize first:

$$\tilde{x}_{ij} = \frac{x_{ij} - \bar{x}_j}{\sqrt{\frac{1}{n}\sum_{i=1}^{n}(x_{ij} - \bar{x}_j)^2}}.$$

Credit data: standardized coefficients shrink smoothly to zero as λ grows.

3.2 Why ridge improves over least squares

Simulation, $n = 50$, $p = 45$, all coefficients nonzero: test MSE is minimized at an intermediate λ, well below least squares.

Theorem: bias and variance of ridge

For $Y = X\beta + \varepsilon$, $\varepsilon \sim (0, \sigma^2 I)$, and $\hat{\beta}_\lambda = (X^\top X + \lambda I)^{-1} X^\top Y$,

$$\mathbb{E}\hat{\beta}_\lambda = (X^\top X + \lambda I)^{-1} X^\top X \beta, \qquad \mathrm{Bias} = -\lambda (X^\top X + \lambda I)^{-1} \beta,$$

$$\mathrm{Var}(\hat{\beta}_\lambda) = \sigma^2 (X^\top X + \lambda I)^{-1} X^\top X (X^\top X + \lambda I)^{-1}, \qquad \mathrm{MSE} = \|\mathrm{Bias}\|^2 + \mathrm{trVar}.$$

Moreover there is always some $\lambda > 0$ with $\mathrm{MSE}(\hat\beta_\lambda) < \mathrm{MSE}(\hat\beta_{\mathrm{OLS}})$ (Hoerl and Kennard, 1970).

Proof [Supplement]. The mean and variance follow from linearity in Y; $(X^\top X+\lambda I)^{-1}X^\top X - I = -\lambda(X^\top X+\lambda I)^{-1}$. With $X^\top X = VD^2V^\top$ and $\theta = V^\top\beta$,

$$\mathrm{MSE}(\lambda) = \sum_{j=1}^{p} \frac{\sigma^2 d_j^2 + \lambda^2\theta_j^2}{(d_j^2+\lambda)^2}.$$

Differentiating, $\mathrm{MSE}'(\lambda) = \sum_j \frac{2d_j^2(\lambda\theta_j^2-\sigma^2)}{(d_j^2+\lambda)^3}$, which equals $-2\sigma^2\sum_j d_j^{-4} < 0$ at $\lambda = 0$, so MSE decreases for small $\lambda > 0$. ■

The gain is largest when some d_j are small (collinearity) or $p \approx n$.

3.3 The lasso

Ridge keeps all p predictors; the lasso

$$\hat\beta^L = \arg\min_\beta\{\mathrm{RSS} + \lambda\|\beta\|_1\}, \qquad \|\beta\|_1 = \sum_{j=1}^{p}|\beta_j|.$$

sets some coefficients **exactly zero** for large λ: variable selection, **sparse** models. Credit data: coefficients enter the path one at a time as λ decreases.

3.4 Constrained formulations and geometry

$$\min_\beta \sum_{i=1}^{n}\Big(y_i - \beta_0 - \sum_j \beta_j x_{ij}\Big)^2 \quad \text{subject to} \quad \sum_j \beta_j^2 \le s \quad \text{resp.} \quad \sum_j |\beta_j| \le s,$$

for some $s = s(\lambda)$ (Lagrange duality); best subset uses $\sum_j \mathbb{I}(\beta_j \neq 0) \le s$.

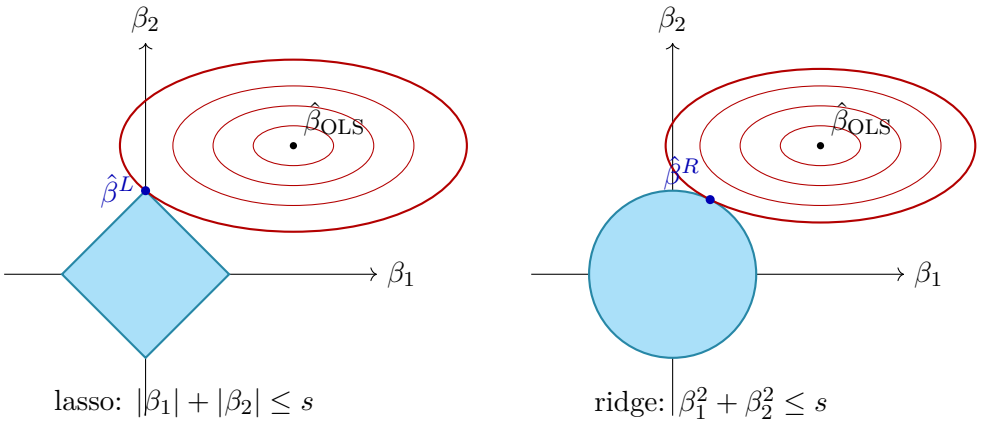


Figure 1: RSS contours and constraint regions. The diamond's corners lie on the axes, so the lasso often hits one ($\hat\beta_1 = 0$ here); the disc has none.

Theorem: orthonormal design

If $n = p$ and $X = I$ (no intercept), then for RSS + λ·penalty:

$$\hat{\beta}_j^R = \frac{y_j}{1+\lambda}, \qquad \hat{\beta}_j^L = \begin{cases} y_j - \lambda/2, & y_j > \lambda/2, \\ y_j + \lambda/2, & y_j < -\lambda/2, \\ 0, & |y_j| \le \lambda/2. \end{cases}$$

Proof [Supplement]. The objective separates: ridge minimizes $(y_j - \beta_j)^2 + \lambda\beta_j^2$; lasso minimizes $(y_j - \beta_j)^2 + \lambda|\beta_j|$, whose subgradient condition $-2(y_j - \beta_j) + \lambda s_j = 0$, $s_j \in \partial|\beta_j|$, gives the soft-thresholding rule. ■

Ridge shrinks proportionally; the lasso soft-thresholds.

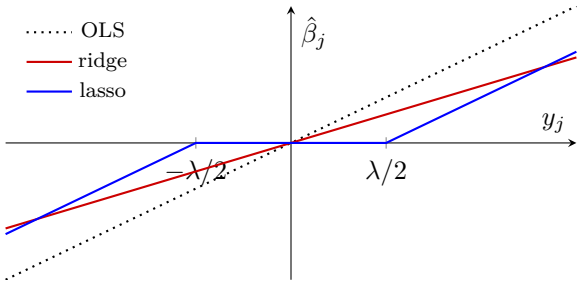


Figure 2: Orthonormal design ($\lambda = 0.6$).

3.5 Dense versus sparse truth

All 45 predictors matter: ridge slightly beats the lasso; only 2 of 45: the lasso wins clearly. The sparsity is unknown, so CV decides.

3.6 Selecting the tuning parameter

Grid of λ, CV error for each, minimizer, refit on all data.

4 Theoretical View in High Dimensions

$Y = X\beta + \varepsilon$, $X \in \mathbb{R}^{n\times p}$, possibly $p \gg n$, where OLS is unstable or not identifiable. **Ridge** assumes β dense but small in ℓ_2; the **lasso** assumes β (approximately) sparse.

Theorem: lasso rate under sparsity (informal)

Let β be s-sparse, $\|\beta\|_0 \le s$. Under a restricted-eigenvalue (RE) condition on X and with $\lambda \asymp \sigma\sqrt{\log p/n}$ (in the normalization $\frac{1}{2n}\text{RSS} + \lambda\|\beta\|_1$),

$$\|\hat{\beta} - \beta\|_2^2 \lesssim \frac{\sigma^2 s \log p}{n}$$

with high probability. Hence consistent estimation is possible when $n \gg s \log p$.

Not knowing the support costs only a $\log p$ factor over the oracle rate $\sigma^2 s/n$.

Why design assumptions are needed. With highly correlated columns, different sparse vectors give similar fits. Guarantees need restricted eigenvalue, compatibility or restricted isometry; exact

support recovery needs an irrepresentable condition and a beta-min condition ($\min_{j\in S}|\beta_j|$ above the noise level).

Ridge does not exploit sparsity but stabilizes estimation under multicollinearity; it suits dense β with small ℓ_2 norm.

Effective complexity. $B_0(s) = \{\beta : \|\beta\|_0 \le s\}$ has complexity about $s \log p$; $B_2(R) = \{\beta : \|\beta\|_2 \le R\}$ is governed by ℓ_2 geometry.

5 Dimension Reduction Methods

5.1 General idea

Construct $M < p$ linear combinations

$$Z_m = \sum_{j=1}^{p} \phi_{mj} X_j, \qquad m = 1, \dots, M,$$

and fit $Y = \theta_0 + \sum_{m=1}^{M} \theta_m Z_m + \varepsilon$ by least squares; the ϕ_{mj} define the method. Since

$$\sum_{m=1}^{M} \theta_m Z_m = \sum_{j=1}^{p} \beta_j X_j, \qquad \beta_j = \sum_{m=1}^{M} \theta_m \phi_{mj},$$

β is constrained to an M-dimensional subspace: less variance, bias if β lies outside it.

5.2 Principal components regression

Directions: principal components, i.e. successive maximal-variance uncorrelated combinations (eigenvectors of the sample covariance).

Example: advertising data, population and ad spending in 100 cities

$Z_1 = 0.839(\texttt{pop} - \overline{\texttt{pop}}) + 0.544(\texttt{ad} - \overline{\texttt{ad}})$, $0.839^2 + 0.544^2 = 1$: the line minimizing squared perpendicular distances. z_{i1} is strongly related to both pop and ad; second-PC scores only weakly.

Assumption: the directions of largest variation in X are those associated with Y. Sparse dependence on many predictors needs many components. $M = p$ is least squares; ridge shrinks low-variance directions smoothly, PCR drops them (Exercise 6.3). Credit data: CV error lowest near $M = 10$, little reduction.

Standardization: generally standardize before PCA, unless all variables share a unit.

Limitation: the directions are *unsupervised*, so they need not be best for predicting Y.

5.3 Partial least squares

PLS chooses $Z_1, \dots, Z_M$ *supervised*: approximating the predictors *and* related to the response.

Theorem: the first PLS direction

With standardized X and centered Y,

$$w_1 = \arg\max_{\|w\|=1} \operatorname{Cov}(Xw, Y)^2 \;\propto\; X^\top Y,$$

i.e. ϕ_{1j} is proportional to the simple-regression coefficient of Y on X_j; $Z_1 = Xw_1$.

Proof [Supplement]. $\mathrm{Cov}(Xw, Y) = \frac{1}{n} w^\top X^\top Y$, and by Cauchy–Schwarz $(w^\top X^\top Y)^2 \le \|w\|^2 \|X^\top Y\|^2$ with equality iff $w \propto X^\top Y$. For standardized X_j the simple-regression slope of Y on X_j is $x_{(j)}^\top Y/n$. ■

Algorithm: PLS (sequential construction)

Standardize X and Y; set $X^{(0)} = X$, $Y^{(0)} = Y$. For $m = 1, \dots, M$:

1. direction $w_m \propto X^{(m-1)\top} Y^{(m-1)}$;
2. component $Z_m = X^{(m-1)} w_m$;
3. regress and update residuals $Y^{(m)} = Y^{(m-1)} - \hat{\theta}_m Z_m$;
4. deflate the predictors $X^{(m)} = X^{(m-1)} - Z_m p_m^\top$, $p_m = X^{(m-1)\top} Z_m / (Z_m^\top Z_m)$ (each column is replaced by its residual after regressing on Z_m).

So $Y = \theta_1 Z_1 + E_1$, $E_1 = \theta_2 Z_2 + E_2$, …; the components are uncorrelated. Advertising: the first PLS direction weights `ad` more than the first PC. In practice PLS is often no better than ridge or PCR: less bias, possibly more variance.

5.4 Theoretical comparison

Method	Assumption on β	Mechanism
lasso	sparsity	ℓ_1 constraint
ridge	small ℓ_2 norm	shrinkage
PLS	low-dimensional structure	iterative projection

With $\Sigma = \mathrm{Cov}(X)$, $c = \mathrm{Cov}(X, Y)$, the m-component PLS estimator lies in the Krylov subspace

$$\mathcal{K}_m(\Sigma, c) = \mathrm{span}\{c, \Sigma c, \dots, \Sigma^{m-1} c\},$$

(least squares restricted to it, i.e. m conjugate-gradient steps on the normal equations); it is consistent if $\beta \in \mathcal{K}_m(\Sigma, c)$. Since $\beta = \Sigma^{-1} c$, this holds e.g. when c lies in the span of m eigenvectors of Σ.

Summary

- Stepwise search: $1 + p(p+1)/2$ models; size by C_p, AIC, BIC, adjusted R^2 or CV.
- Ridge shrinks (beats OLS for some λ); the lasso shrinks and selects ($s \log p/n$ under design conditions).
- PCR: unsupervised components; PLS: supervised directions (Krylov subspace).

6 Selected exercises

Motivation. ISLR Ex. 6.5 and 6.7 (duplicated predictors, Bayesian view) and ridge in the singular-value basis.

Exercise 6.1 (Ridge and lasso with duplicated predictors; ISLR Ex. 6.5)

Let $n = p = 2$, $x_{11} = x_{12} = a$, $x_{21} = x_{22} = -a$ with $a \neq 0$, and $y_1 + y_2 = 0$, so that the intercept is 0 and is omitted. Let $\lambda > 0$.
(a) Write the ridge problem and show that its solution has $\hat\beta_1 = \hat\beta_2$.
(b) Write the lasso problem and show that its solution is in general not unique; describe all solutions.

Solution. With $s = \beta_1 + \beta_2$ the residuals are $y_1 - as$ and $y_2 + as = -(y_1 - as)$, so $\mathrm{RSS} = 2(y_1 - as)^2$.

(a) Ridge minimizes $R(\beta) = 2(y_1 - a(\beta_1 + \beta_2))^2 + \lambda(\beta_1^2 + \beta_2^2)$.

Step 1. R is strictly convex (a convex quadratic plus the strictly convex $\lambda\|\beta\|^2$) and tends to ∞ as $\|\beta\| \to \infty$, so it has a unique minimizer.

Step 2. $R(\beta_1, \beta_2) = R(\beta_2, \beta_1)$, so swapping the coordinates of the minimizer gives a minimizer; by uniqueness $\hat\beta_1 = \hat\beta_2 = b$. Minimizing $2(y_1 - 2ab)^2 + 2\lambda b^2$ gives $b = 2ay_1/(4a^2 + \lambda)$.

(b) The lasso minimizes $L(\beta) = 2(y_1 - a(\beta_1 + \beta_2))^2 + \lambda(|\beta_1| + |\beta_2|)$.

Step 1. $|\beta_1| + |\beta_2| \geq |s|$, with equality iff $\beta_1\beta_2 \geq 0$. Hence $L(\beta) \geq q(s) := 2a^2(s - m)^2 + \lambda|s|$ with $m = y_1/a$, and equality holds iff β_1, β_2 have the same sign (or one is 0).

Step 2. q is strictly convex with unique minimizer $\hat s = \mathrm{sign}(m)\big(|m| - \lambda/(4a^2)\big)_+$ (the soft-thresholding computation of the orthonormal-design theorem, with $4a^2$ in place of 2).

Step 3. For every β, $L(\beta) \geq q(s) \geq q(\hat s)$, and $L(\beta) = q(\hat s)$ iff $\beta_1 + \beta_2 = \hat s$ and $\beta_1\beta_2 \geq 0$. So the lasso solutions are

$$\{(\beta_1, \beta_2) : \ \beta_1 + \beta_2 = \hat s, \ \beta_1\beta_2 \geq 0\},$$

the segment from $(\hat s, 0)$ to $(0, \hat s)$: infinitely many if $\hat s \neq 0$, only $(0, 0)$ if $|y_1| \leq \lambda/(4|a|)$. ■

Exercise 6.2 (Bayesian view of ridge and lasso; ISLR Ex. 6.7)

Let $y_i = \sum_{j=1}^p x_{ij}\beta_j + \varepsilon_i$ with $\varepsilon_i \overset{\text{iid}}{\sim} N(0, \sigma^2)$ (centered data, σ^2 known).
(a) Write the likelihood.
(b) With independent double-exponential priors $p(\beta_j) = \frac{1}{2b}e^{-|\beta_j|/b}$, write the posterior and show that its mode is the lasso estimate for some λ.
(c) With independent $N(0, c)$ priors, show that the posterior is normal and that its mode and its mean both equal the ridge estimate for some λ.

Solution.

(a) $L(\beta) = (2\pi\sigma^2)^{-n/2}\exp\{-\|y - X\beta\|^2/(2\sigma^2)\}$.

(b) The posterior density is proportional to $L(\beta)\prod_j p(\beta_j)$:

$$p(\beta \mid y) \propto \exp\Big\{-\frac{1}{2\sigma^2}\Big(\|y - X\beta\|^2 + \frac{2\sigma^2}{b}\|\beta\|_1\Big)\Big\}.$$

As $u \mapsto e^{-u/(2\sigma^2)}$ is strictly decreasing, the mode minimizes $\mathrm{RSS} + \lambda\|\beta\|_1$, $\lambda = 2\sigma^2/b$.

(c)

Step 1. Here $p(\beta \mid y) \propto \exp\{-Q(\beta)/(2\sigma^2)\}$ with $Q(\beta) = \|y - X\beta\|^2 + \lambda\|\beta\|^2$, $\lambda = \sigma^2/c$.

Step 2. With $A = X^\top X + \lambda I \succ 0$ and $m = A^{-1}X^\top y$, using $m^\top A = y^\top X$,

$$Q(\beta) = \beta^\top A\beta - 2\beta^\top X^\top y + y^\top y = (\beta - m)^\top A(\beta - m) + y^\top y - m^\top Am.$$

Step 3. So $p(\beta \mid y) \propto \exp\{-\frac{1}{2}(\beta - m)^\top(\sigma^2A^{-1})^{-1}(\beta - m)\}$, the $N(m, \sigma^2A^{-1})$ density, whose mean and mode are $m = (X^\top X + \lambda I)^{-1}X^\top y$, the ridge estimate. ■

Exercise 6.3 (Ridge in the singular-value basis)

Let $X = \sum_{j=1}^{r} d_j u_j v_j^\top$ be a singular value decomposition ($d_j > 0$, $r = \mathrm{rank} X$, orthonormal u_j and v_j), $\lambda > 0$, $\hat\beta_\lambda = (X^\top X + \lambda I)^{-1} X^\top y$ and $H_\lambda = X(X^\top X + \lambda I)^{-1} X^\top$.

(a) Show that $\hat\beta_\lambda = \sum_{j=1}^{r} \dfrac{d_j\, u_j^\top y}{d_j^2 + \lambda}\, v_j$.

(b) Show that $\|\hat\beta_\lambda\|$ is nonincreasing in λ, strictly decreasing unless $X^\top y = 0$, and tends to 0.

(c) Show that $\mathrm{df}(\lambda) = \mathrm{tr} H_\lambda = \sum_{j=1}^{r} d_j^2/(d_j^2 + \lambda)$, decreasing from r (as $\lambda \to 0$) to 0.

Solution [Supplement].

(a) Extend $v_1, \ldots, v_r$ to an orthonormal basis $v_1, \ldots, v_p$ of $\mathbb{R}^p$. Then $(X^\top X + \lambda I) v_j = (d_j^2 + \lambda) v_j$ for $j \le r$ and $= \lambda v_j$ for $j > r$, so $(X^\top X + \lambda I)^{-1} v_j = v_j/(d_j^2 + \lambda)$ for $j \le r$. Since $X^\top y = \sum_{j \le r} d_j (u_j^\top y) v_j$, the formula follows.

(b) By orthonormality,

$$\|\hat\beta_\lambda\|^2 = \sum_{j=1}^{r} \frac{d_j^2 (u_j^\top y)^2}{(d_j^2 + \lambda)^2},$$

each term nonincreasing in λ and strictly decreasing when $u_j^\top y \neq 0$. Since $X^\top y = \sum_j d_j (u_j^\top y) v_j$, all $u_j^\top y = 0$ iff $X^\top y = 0$. Each term tends to 0 as $\lambda \to \infty$.

(c) $H_\lambda = X\hat\beta_\lambda$ as a map of y, i.e. $H_\lambda y = \sum_j \frac{d_j^2}{d_j^2+\lambda}(u_j^\top y) u_j$, so $H_\lambda = \sum_j \frac{d_j^2}{d_j^2+\lambda} u_j u_j^\top$ and $\mathrm{tr} H_\lambda = \sum_j \frac{d_j^2}{d_j^2+\lambda}$ (as $\mathrm{tr} u_j u_j^\top = 1$). Each summand decreases from 1 to 0.

The coordinate along u_j is shrunk by $d_j^2/(d_j^2 + \lambda)$, most for low-variance principal components (PCR keeps or drops them). ■